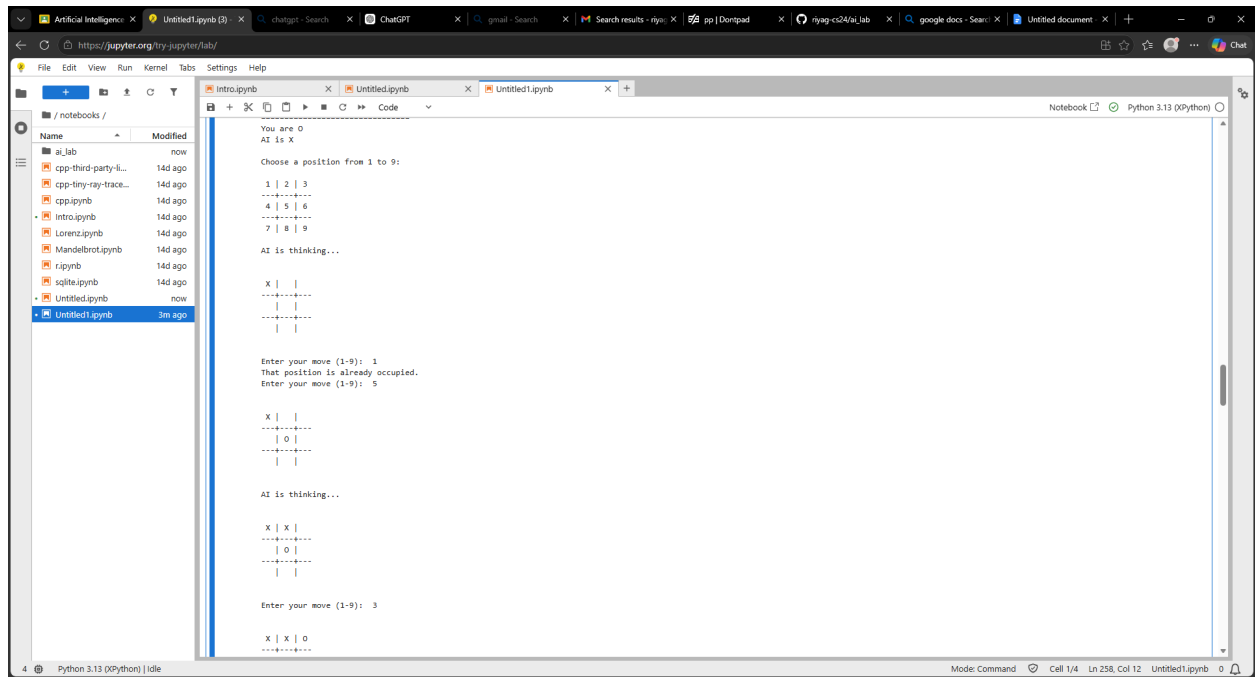


output:(tic-tac-toe)



```
File Edit View Run Kernel Tabs Settings Help
/notebooks /
Name Modified
ai_lab now
cpp-third-party-li... 14d ago
cpp-tiny-ray-trace... 14d ago
cpp.ipynb 14d ago
Intro.ipynb 14d ago
Lorenz.ipynb 14d ago
Mandelbrot.ipynb 14d ago
ripiynb 14d ago
sqlite.ipynb 14d ago
Untitled.ipynb now
Untitled1.ipynb 3m ago

Intro.ipynb X Untitled.ipynb X Untitled1.ipynb X
+ + +
Notebook Python 3.13 (xPython)

You are O
AI is X

Choose a position from 1 to 9:

1 | 2 | 3
-----
4 | 5 | 6
-----
7 | 8 | 9

AI is thinking...

X | | 
-----
| | 
-----
| | 

Enter your move (1-9): 1
That position is already occupied.
Enter your move (1-9): 5

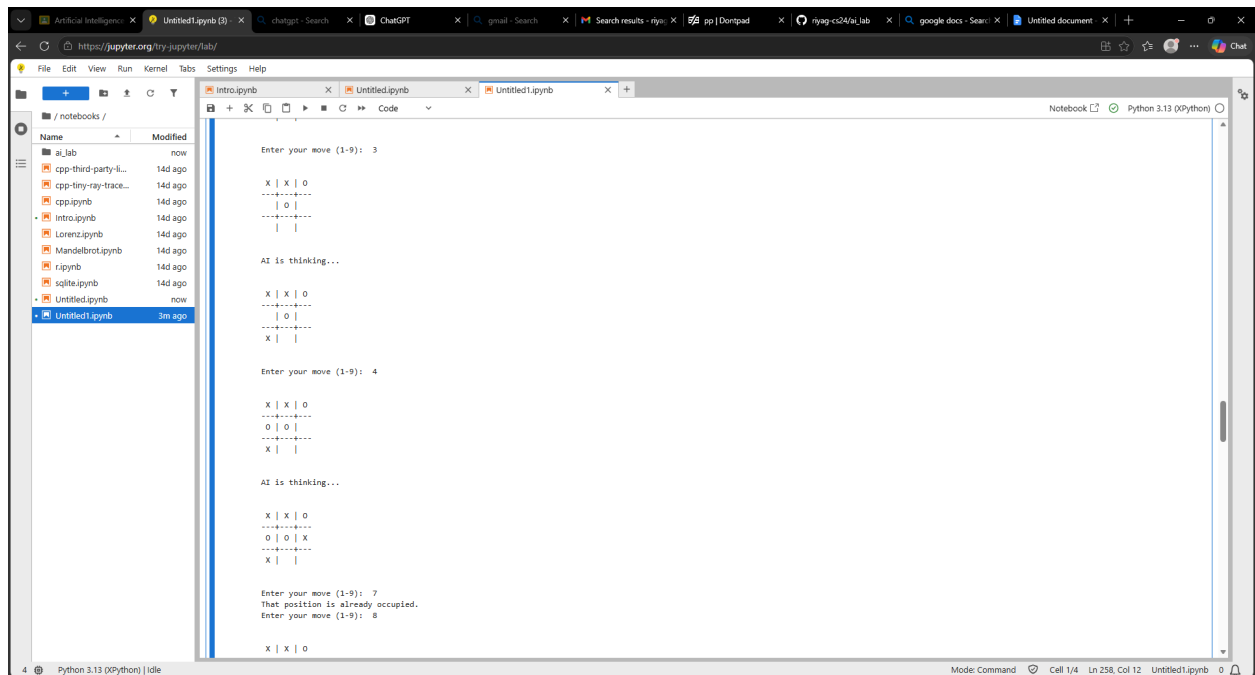
X | | 
-----
| 0 | 
-----
| | 

AI is thinking...

X | X | 
-----
| 0 | 
-----
| | 

Enter your move (1-9): 3

X | X | 0
-----
```



```
File Edit View Run Kernel Tabs Settings Help
/notebooks /
Name Modified
ai_lab now
cpp-third-party-li... 14d ago
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cpp.ipynb 14d ago
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Untitled.ipynb now
Untitled1.ipynb 3m ago

Intro.ipynb X Untitled.ipynb X Untitled1.ipynb X
+ + +
Notebook Python 3.13 (xPython)

Enter your move (1-9): 3

X | X | 0
-----
| 0 | 
-----
| | 

AI is thinking...

X | X | 0
-----
| 0 | 
-----
X | | 

Enter your move (1-9): 4

X | X | 0
-----
0 | 0 | 
-----
X | | 

AI is thinking...

X | X | 0
-----
0 | 0 | X
-----
X | | 

Enter your move (1-9): 7
That position is already occupied.
Enter your move (1-9): 8

X | X | 0
-----
```

The screenshot shows a Jupyter Notebook with a file explorer on the left listing various notebooks. The main area contains a game state represented by a 3x3 grid:

```
X | X | O
-----
O | O | X
-----
X | |
```

Below the grid, there is a text prompt: "Enter your move (1-9): 7". A message follows: "That position is already occupied." Another prompt appears: "Enter your move (1-9): 8".

The code cell at the bottom contains the following Python code:

```
[3]: # =====
# VACUUM CLEANER PROBLEM
# User Input Version
# =====

def display_state(state):
    vacuum, room_A, room_B = state

    print("Un-----")
    print("Current State")
    print("-----")
```

output(vaccum cleaner)

The screenshot shows the same Jupyter Notebook interface, but now displaying the output of the vacuum cleaner simulation. The code cell contains the following Python code:

```
# Create Initial state
initial_state = (vacuum, room_A, room_B)

# Run the vacuum cleaner
vacuum_cleaner(initial_state)

=====
VACUUM CLEANER PROBLEM
=====
Enter vacuum position (A/B): a
Is Room A dirty? (yes/no): yes
Is Room B dirty? (yes/no): no

Initial State:
-----
Current State
-----
Vacuum Position : A
Room A          : Dirty
Room B          : Clean
-----
Step 1 : SUCK
-----
Current State
-----
Vacuum Position : A
Room A          : Clean
Room B          : Clean
-----

GOAL REACHED!
Both rooms are clean.
=====
```

The output shows the initial state, the user input for vacuum position (A/B) and room dirtiness, the initial state, the current state, the vacuum position, and the goal reached message.